

Create an enterprise level QNA agent prototype. Generate datasets with overlapping entities for FMCG company. Create document involving same entities. Multiple dataset and documents should have overlapping theme and entities. Based on these create an agent with sub-agents which should be doing,

- Structured data retrieval
- Unstructured data retrieval
- Internet search
- Coding

The main agent should be synthesizing answering from sub-agents and should answer a question.

Here are some of the attributes which should be present in the agent,

- Support single-turn and multi-turn conversational interactions.
- Support greeting, capability introduction, and out-of-scope request handling.
- Support intent validation before data retrieval.
- Support clarification for ambiguous or incomplete user requests.
- Support contextual follow-up questions by maintaining conversation history.
- Support semantic understanding, including aliases, abbreviations, and typo correction.
- Support multilingual and mixed-language queries.
- Support conversation context preservation across interactions.
- Support secure access with SQL safety controls.
- Support structured and unstructured data retrieval from multiple data sources.
- Support document retrieval with source citations.
- Support hybrid data retrieval
- Support answer validation, retry mechanisms, and response quality evaluation.
- Support standardized formatting, including markdown tables and unit-aware presentation.
- Support temporal reasoning for current, historical, and comparative period analysis.
- Support context-aware follow-up suggestions within supported business domains.
- Support conversation memory optimization for long-running sessions.
- Support metadata queries.
- Support multiple KPIs, entities, dimensions, and hierarchical business structures.
- Support analytical comparisons across KPIs, entities, periods, and business domains.
- Support hierarchy-aware fallback for unsupported entities or granularities.
- Support metadata discovery for available KPIs, dimensions, periods, and datasets.
- Support document filtering using metadata, tags, and recency.
- Support transparent reporting of assumptions, data availability, and system limitations.
- Support graceful handling of unsupported or unavailable requests.

Create a GitHub repository and share the link. Document all the design decision. Feel free to use AI coding tools. The final submission should have a README.md and docs folder which explains the inner working for the agent, and a notebook with prerun sample of test questions and responses which have been generated by your agent. Build a point of view on cost, latency, model usage and document trade off and key design decisions. Deadline send the submission within 7 days with the link of the github repository.